

HETEROGENEITY WITH COMPLETE MARKETS

Part I: Theory

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Motivation

- Any economic model is deliberately an abstraction from reality.
- For a long time, **macroeconomics** abstracted from differences across economic agents, focusing on **representative-agent models**.
- Some of these differences were safe to ignore (eye color), some were not obviously safe to ignore (preferences, income, wealth, etc.), but there were good reasons for that.
- Our job today is to find exactly when the representative-agent formulation is valid and see how **fragile** that validity is.

Plan for this lecture

- Does heterogeneity matter for aggregates? When markets are **complete**, it might not.
- Two results:
 - **Exact aggregation** (Gorman, 1961)
 - **Equilibrium-point aggregation** (Constantinides, 1982)
- Punchline: aggregation is a **knife-edge property** of **complete markets**.

Warm-up: a static pure exchange economy

- I consumers with identical homothetic preferences over N goods.
- Consumer i 's endowment is $y^i \in \mathbb{R}_+^N$; prices are $\mathbf{p} \in \mathbb{R}_+^N$; a consumption plan is $\mathbf{c} = (c^1, \dots, c^I)$ with $c^i \in \mathbb{R}_+^N$ for every i .
- Consumer i solves:

$$\max_{\mathbf{c}} U(\mathbf{c}) \quad \text{s.t.} \quad \mathbf{p}\mathbf{c} \leq \mathbf{p}y^i =: W^i \quad (\text{UM})$$

- Markets clear:

$$\sum_i c^i = \sum_i y^i \quad (\text{MC})$$

Warm-up: competitive equilibrium

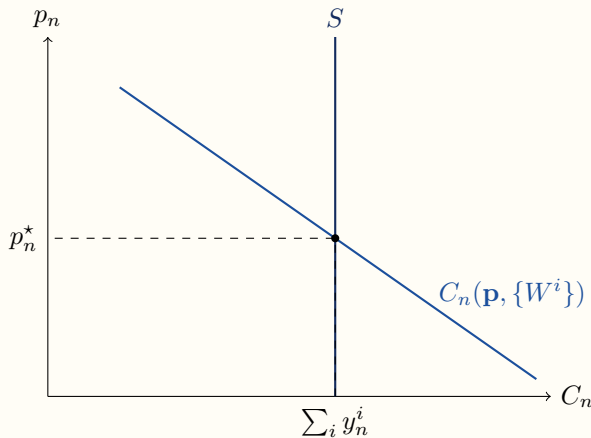
Competitive equilibrium

A *competitive equilibrium* is a pair (c, p) such that:

1. c^i solves consumer i 's problem (UM) given prices p , for every i .
2. Markets clear ((MC) holds).

- Agents differ only in their endowments — does that matter?
- If we **reshuffle** endowments across agents, **does the equilibrium change?**

Does equilibrium change if we reshuffle?



- Fix attention on the market for **good n** , and let $\{W^i\}$ denote the wealth distribution.
- Supply is **perfectly inelastic** in a pure exchange economy.
- Aggregate demand:
 $C_n(\mathbf{p}, \{W^i\}) = \sum_i c_n^*(\mathbf{p}, W^i)$.
- Reshuffle $\{W^i\}$ holding $\sum_i W^i$ fixed: does the **price** change?

Homotheticity makes demand linear in wealth

Claim: Demand is **homogeneous of degree 1** in wealth.

Proof.

- (1) Let $c^*(\mathbf{p}, W^i)$ solve (UM). Then $\lambda c^*(\mathbf{p}, W^i)$ is affordable at wealth λW^i .
- (2) Any c affordable at λW^i has c/λ affordable at W^i , so $c^*(\mathbf{p}, W^i) \succeq c/\lambda$.
- (3) Homotheticity implies $\lambda c^*(\mathbf{p}, W^i) \succeq c$.
- (4) Hence $\lambda c^*(\mathbf{p}, W^i)$ solves the problem at λW^i : $c^*(\mathbf{p}, \lambda W^i) = \lambda c^*(\mathbf{p}, W^i)$.



Exact aggregation

- This means that we can write $c^*(\mathbf{p}, W^i) = \frac{W^i}{W} c^*(\mathbf{p}, W)$, where $W = \sum_i W^i$.

- Summing across agents:

$$C(\mathbf{p}, \{W^i\}) = \sum_i c^*(\mathbf{p}, W^i) = c^*(\mathbf{p}, W).$$

- Aggregate demand equals that of **one consumer** who owns all the wealth.
- This result, due to *Gorman (1961)*, carries over to more familiar macro models as long as markets are complete.

Arrow–Debreu economy with endowment shocks

Environment

- Time is discrete and infinite.
- Single non-storable good.
- I infinitely-lived consumers $i \in \{1, \dots, I\}$.
- Each agent's endowment y_t^i is stochastic:
 - Not necessarily Markov.
 - There may be **aggregate uncertainty**.
- Markets are complete.

Market structure

- The **state** of the economy is $s_t = \{y_t^1, \dots, y_t^I\}$:
 - Two states with the same aggregate but different distributions are, formally, **different states**: $(y^1, y^2) = (3, 1)$ and $(2, 2)$ are not the same s_t .
- History up to t is $s^t = \{s_0, \dots, s_t\}$ and occurs with probability $\pi(s^t)$.
- There is a complete set of contingent claims on histories s^t all trading at $t = 0$.
- Price system $\mathbf{p} = \{p_t(s^t)\}$: price of one unit of consumption delivered at t if history s^t occurs.

Preferences and market clearing

- Preferences are identical and time-separable:

$$U(c^i) = \sum_t \sum_{s^t} \beta^t \pi(s^t) u(c_t^i(s^t))$$

- Consumer i solves:

$$\max_{c^i} U(c^i) \quad \text{s.t.} \quad \sum_t \sum_{s^t} p_t(s^t) c_t^i(s^t) \leq \sum_t \sum_{s^t} p_t(s^t) y_t^i(s^t) \quad (\text{AD-UM})$$

- Market clearing:

$$\sum_i c_t^i(s^t) \leq \sum_i y_t^i(s^t) \quad \text{for all } t, s^t \quad (\text{AD-MC})$$

Arrow–Debreu competitive equilibrium

AD-CE

An Arrow–Debreu competitive equilibrium consists of a price system and consumption plan such that

1. Individual consumption plans solve (AD-UM) given the price system.
2. Markets clear ((AD-MC)).

- We can ask ourselves the **same question** as before:
 - do differences across agents matter?
 - does the equilibrium depend on these differences?
- We will obtain the **same answer**.

Homotheticity again delivers aggregation

- Define time-0 wealth: $W_0^i = \sum_t \sum_{s^t} p_t(s^t) y_t^i(s^t)$.
- Assume $u(c_t) = \frac{c_t^{1-\sigma} - 1}{1-\sigma}$.
 - Preferences are homothetic, even in the log case.
- The consumer's problem can be written as

$$\max_c U(c) \quad \text{s.t.} \quad \mathbf{p} \cdot c \leq W_0^i,$$

where $\mathbf{p} \cdot c$ is the inner product of prices and the consumption plan.

Claim: Demand is **homogeneous of degree 1** in wealth.

Effect of redistribution

- FOCs deliver **linear wealth-expansion paths**:

$$c_t^i(s^t) = b_t(\mathbf{p}, s^t) W_0^i$$

where $b_t(\mathbf{p}, s^t)$ does *not* depend on i .

- Consider a **redistribution of endowments**:

$$\left\{ \{dy_t^i(s^t)\} : \sum_i dy_t^i(s^t) = 0 \text{ for all } t, s^t \right\}$$

- Then, aggregate demand is **invariant to redistribution**:

$$dC_t(s^t) = \sum_i b_t(\mathbf{p}, s^t) dW_0^i = 0$$

Result: exact aggregation

- The heterogeneous-agent economy and a single-consumer economy endowed with $W_0 = \sum_i W_0^i$ deliver **identical** prices and aggregates — this is **Gorman aggregation**.
- Remark: the result is useful but **fragile**. It need not survive
 - the loss of complete markets;
 - other forms of heterogeneity (e.g., in preferences).

The same logic goes further — as exercises

- The exact-aggregation argument extends unchanged to a **production economy** with heterogeneous initial capital k_0^i and a CRS firm (**PS1, Ex. 1a**).
- Add a subsistence level $\bar{c} > 0$: preferences are no longer homothetic, but Engel curves stay linear with a common slope, so a **representative consumer still exists** (**PS1, Ex. 1b**).

How crucial is market completeness?

Imposing a borrowing constraint

- Let's make the markets incomplete by imposing a **no-borrowing constraint**:
 $c^i(s^t) \leq y^i(s^t)$ for every i, t, s^t .
- Since all contingent claims are in zero net supply and nobody can go short, **autarky** is the only feasible allocation that can clear the markets.
- Who prices the state-contingent claims?

Maximum willingness to pay

- Evaluate agent i 's Euler equation at *autarky*:

$$\text{MRS}_i(s^t) = \beta^t \pi(s^t) \frac{u'_i(y_t^i(s^t))}{u'_i(y_0^i(s_0))}$$

- This is exactly his maximum willingness to pay for a claim that pays at s^t :
 - If $p_t(s^t)$ is below this, agent i would **save** (move resources to s^t).
 - If $p_t(s^t)$ is above this, agent i would **borrow** (bring resources to s_0), but can't.

The equilibrium price

- For autarky to be an equilibrium, the price must be high enough that **nobody wants to buy the claim**:

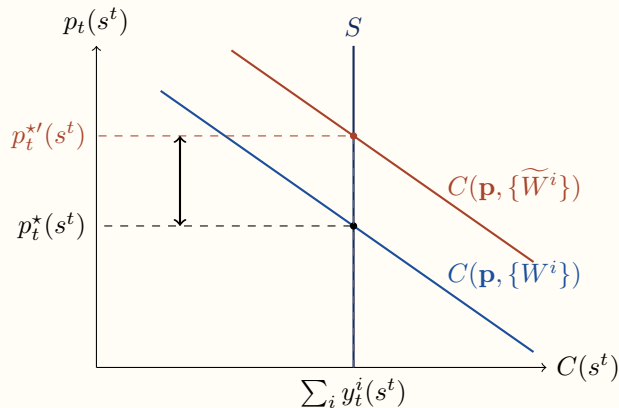
$$p_t(s^t) = \max_i \text{MRS}_i(s^t).$$

- Hence $p_t(s^t)$ is set by whoever has the single highest willingness to pay.
- In equilibrium, for each history, the price is set by the agent with the strongest desire to save into s^t . **It is not a single agent doing all the pricing.**

What redistribution does

- When markets are incomplete, we can find a **redistribution** that **changes prices**:
 - Take the agent setting $p_t(s^t)$, shift some of *his* endowment in s^t to someone else.
 - His willingness to pay changes and **the price moves with it**.
- The equilibrium depends on differences across agents: **heterogeneity matters!**

How large is the effect?



- Redistribution **moves** the demand curve: **aggregation breaks down**.
- Important question: **are these changes quantitatively large?**

Heterogeneity in preferences

Exact aggregation is fragile

- Everything so far has relied on **identical, homothetic** preferences, with heterogeneity only in wealth.
- Add an idiosyncratic shock to preferences, and exact aggregation **breaks down** again:
 - We can't give all wealth to a single agent and expect him to make the same choices as in the aggregate (why?).
- Not all is lost: we can still construct a *fictitious* representative agent — one that ranks aggregates the same way the competitive equilibrium does, even though no individual looks like it.

Setup

- Same exchange economy, but the idiosyncratic shock is now two-dimensional: $s_t^i = (y_t^i, \phi_t^i)$.
- y_t^i : stochastic endowment (as before). ϕ_t^i : shock to **preferences**.
- Consumer i solves:

$$\max_{c^i} \sum_t \sum_{s^t} \beta^t \pi(s^t) u(c_t^i(s^t), \phi_t^i) \quad \text{s.t.} \quad \sum_t \sum_{s^t} p_t(s^t) c_t^i(s^t) \leq \sum_t \sum_{s^t} p_t(s^t) y_t^i$$

- Markets remain complete.

Finding the equilibrium using welfare theorems

- The definition of competitive equilibrium is standard, and with complete markets, the CE allocation is **Pareto efficient**.
- Every Pareto-efficient allocation solves a planner's problem for *some* Pareto weights a :

$$\max_{\{c^i\}} \sum_i a^i \sum_t \sum_{s^t} \beta^t \pi(s^t) u(c_t^i(s^t), \phi_t^i) \quad \text{s.t.} \quad \sum_i c_t^i(s^t) \leq \sum_i y_t^i$$

- Negishi–Mantel algorithm: find the a that delivers the CE allocation for a *given* initial distribution of resources.
- We will take the a as given, but we must remember that it has encoded a distribution of wealth across agents.

Splitting the planner's problem

- The planner's problem splits into two stages:

1. **Cross-sectional**: given aggregate C_t , how is it shared out?

$$U(C_t, a, \{\phi_t\}) = \left\{ \max_{\{c^i\}} \sum_i a^i u(c^i, \phi^i) \quad \text{s.t.} \quad \sum_i c^i = C_t \right\}$$

2. **Intertemporal**: allocate C_t over time using $U(\cdot)$ as **social preferences**:

$$\max \sum_t \sum_{s^t} \beta^t \pi(s^t) U(C_t(s^t), a, \{\phi_t\}) \quad \text{s.t.} \quad C_t(s^t) \leq \sum_i y_t^i$$

- If $U(C_t, a, \{\phi_t\})$ has a closed form, we can treat it as the **preferences of the representative consumer**.

A worked example: CRRA with multiplicative shocks

- Let $u(c, \phi) = \phi \frac{c^{1-\sigma}}{1-\sigma}$. Solving the cross-sectional stage explicitly:

$$U(C_t, a, \{\phi_t\}) = \Phi_t \frac{C_t^{1-\sigma}}{1-\sigma}, \quad \Phi_t = \left(\sum_i (a^i \phi_t^i)^{1/\sigma} \right)^\sigma$$

- The functional form of U is **the same** as the individual's — the economy behaves *as if* there were a representative CRRA consumer facing a preference shock Φ_t .
- This is **equilibrium-point aggregation** (*Constantinides, 1982*): unlike Gorman aggregation, U depends on a — i.e., **on the wealth distribution**.

Two special cases

- Turn off **idiosyncratic shocks** ($\phi_t^i \equiv 1$): Φ_t becomes a constant, and we are back to exact aggregation.
- Turn off **wealth** heterogeneity instead ($a^i = 1$ for all i): Φ_t still depends on the cross-sectional **dispersion** of ϕ_t^i .
- The same happens with **skill** shocks: see the application below and **PS1, Exercise 2**.

Application: business cycle accounting

Business cycle accounting

- One way to make an RA growth model fit *any* data: let **wedges** absorb the gap. Wedges are deviations of the data from what a frictionless economy's own equilibrium conditions would require.
- E.g., on the labor-leisure margin, a frictionless economy sets $MRS_t = MRT_t$. The **labor wedge** $\tau_{n,t}$ is the gap between them: $MRS_t = (1 - \tau_{n,t}) MRT_t$ — analogous wedges sit on the production, investment, and resource-constraint margins.
- *Chari, Kehoe and McGrattan (2007)* **formalized** this into business cycle accounting: measure the wedges, feed them back into the model, and see which account for the fluctuations. The aim is to guide researchers “to focus on the **key margins** that need to be distorted.”

But there is a catch

- The **tempting** reading: the dominant wedge stands in for a friction **in that market** — a big labor wedge, build a labor-market friction.
- **CKM's own caveat**: a wedge only tells you where the symptom shows up — the true friction can exist somewhere else entirely.
- It can be worse: a wedge may reflect **no friction at all**. **Differences across agents** can shift the representative agent's preferences (recall Φ_t), its technology, or the mapping from hours to effective labor, and a standard RA model reads all of these as wedges.
- Let's see this in the simplest possible example.

A minimal example

- Two skill types, equal Pareto weight, $u(c, n) = \ln c + \ln(1 - n)$, linear technology $Y_t = H_t = \sum_i \theta_t^i n_t^i$.
- Mean-preserving dispersion: $\theta_t^1 = 1 + \varepsilon_t$, $\theta_t^2 = 1 - \varepsilon_t$, so $\bar{\theta}_t = 1$ *always*: no technology shock, ever, by construction.
- The only thing that moves is ε_t : how spread out skills are in the cross section.

What actually happens

- Solving the planner's problem: $c^1 = c^2 = c^* = \frac{1}{2}$. Consumption is **fully insured**, constant in ε_t .
- Aggregate (efficiency-weighted) output $Y_t = H_t = 2c^* = 1$ is also constant in ε_t .
- But *physical* hours $N_t = n_t^1 + n_t^2$ fall as dispersion rises: the low-skill agent works less, and it is not offset one-for-one in physical units.
- Nothing is inefficient. Nothing is friction-bound. Only ε_t moved.

Through the lens of an RA model

- An outside observer sees C_t , Y_t , and *physical* hours N_t , not θ_t^i and not ε_t .
- The observer fits a *per-capita* RA model: $\ln \bar{c}_t + \ln(1 - \bar{n}_t)$ and $\bar{y}_t = A_t \bar{n}_t$, with $\bar{c}_t = C_t/2$, $\bar{y}_t = Y_t/2$, $\bar{n}_t = N_t/2$.
- Measured TFP $A_t = Y_t/N_t$ and the labor wedge, $\frac{\bar{c}_t}{1 - \bar{n}_t} = (1 - \tau_{n,t}) A_t$, both move with ε_t .

	$\varepsilon = 0$	$\varepsilon = 0.2$
$C = Y$	1.000	1.000
N	1.000	0.958
Measured TFP A	1.000	1.043
Labor wedge τ_n	0	0.080

The punchline

- A $\sim 4\%$ “TFP shock” and an *exact* 8% “labor market distortion,” with **zero** technology change, **zero** friction, **zero** policy, under **complete markets**: all that happened is that the distribution of skill got wider.
- The lesson: complete markets deliver a representative agent, but **not the one you would write down**. Heterogeneity does not wash out in the aggregate: it shows up as the representative agent’s **shocks**.
- This is not a flaw in business-cycle accounting. It is **exactly the caveat CKM themselves flagged**: a wedge is not, by itself, evidence of *where* a friction lives — or whether there is a friction at all.
- In a calibrated growth model, idiosyncratic skill shocks can make the correlation between hours and labor productivity weak or even negative (*Maliar and Maliar, 2003*).

Summary

- Complete markets + homothetic, identical preferences, heterogeneity only in wealth $\Rightarrow$ **exact** aggregation: the RA is the average agent, full stop.
- Add idiosyncratic preference/skill shocks $\Rightarrow$ **equilibrium-point** aggregation: a fictitious RA exists, but depends on the distribution of shocks and wealth.
- Either way, *complete markets* is doing all the work: remove it (e.g., a no-borrowing constraint) and **redistribution moves prices**.
- Next: are markets complete?

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